AI for All, or AI for Ready Schools?

Implementation Capacity, Early-Warning Support Need, and Public Monitoring in Hong Kong Secondary Schools

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Contents

Executive Summary

This report examines whether Hong Kong secondary schools show uneven publicly observable capacity to absorb and implement AI education policy. The analysis is framed as a politics and public administration study: the central concern is not whether schools are good or bad, but whether equal funding can become equal student opportunity when schools differ in planning capacity, digital infrastructure, teacher development, governance, and public reporting practices.

The project uses a support-prioritisation index rather than a ranking index. The Foundational AI Implementation Capacity Index captures visible conditions that may help schools absorb future AI funding. The Early AI-Specific Signal Index captures direct AI-related public evidence, but it is interpreted cautiously because the funding programme had not yet rolled out at school level in the collected evidence. The Early-Warning Support Need Index identifies candidates for validation and support rather than schools to blame.

The evidence base covers 441 secondary schools. The crawler collected public website evidence for 440 school websites, including 8,231 HTML pages and 1,931 PDFs. After text extraction, 8,763 documents contained usable text. The final analytical dataset retained 12,997 cleaned evidence items after deduplication and confidence weighting.

Headline results

Mean foundational capacity is 31.01/100 and mean early AI-specific signal is 20.91/100. The median early AI-specific signal is 12.50/100, meaning direct AI evidence is still thin for many schools. Mean early-warning support need is 74.28/100, while publication visibility is strongly associated with index values. These results support tiered support and manual validation, not public ranking.

The main policy implication is that AI for All requires Capacity for All. A one-off school grant can help, but it will not automatically produce equal implementation. Public monitoring should therefore combine a small standardised reporting system, support tiers, teacher development, governance guidance, infrastructure support, and manual validation of low-visibility cases.

1. Introduction

Artificial intelligence is becoming a major education policy priority in Hong Kong. The policy agenda is not only about buying software. It requires school planning, teacher confidence, digital infrastructure, curriculum integration, student learning opportunities, procurement decisions, and responsible use safeguards. That makes AI education a public administration problem: policy outcomes depend on whether schools can convert central support into local implementation.

Hong Kong's AI education agenda is moving quickly. The 2025 Policy Address announced HK$2 billion under the Quality Education Fund to support digital education, including AI literacy, AI education in the core curriculum, and teacher training. EDB Circular Memorandum No. 221/2025 then introduced the AI for Empowering Learning and Teaching Funding Programme, under which eligible publicly funded schools may receive a one-off HK$500,000 grant.

The project asks a practical question: which schools may need more validation and support before funding can be converted into meaningful AI-assisted teaching and student opportunity? The analysis does not claim to measure actual classroom practice. It measures publicly observable evidence and treats low evidence as a candidate signal for follow-up, especially when publication visibility is also low.

- How much foundational capacity is visible in public evidence?
- How much direct AI-specific activity is visible before policy rollout?
- Which dimensions of implementation capacity appear weakest?
- Do support needs appear uneven across districts and school characteristics?
- How should policymakers use the evidence without creating school rankings?

2. Policy and Literature Framework

2.1 Policy implementation capacity

Policy implementation theory emphasises that policy outputs do not automatically become policy outcomes. Schools are frontline organisations. They translate policy into plans, training, procurement, curriculum resources, classroom practice, and monitoring routines. This report therefore treats the AI funding programme as an organisational implementation challenge rather than a simple technology adoption problem.

2.2 Digital inequality and capability

Digital inequality is not only about whether a school has devices. Meaningful use depends on infrastructure, human capacity, social support, institutional routines, and the ability to convert resources into valued opportunities. The capability approach is useful here: equal inputs do not guarantee equal capabilities. A school with strong planning and teacher development may turn HK$500,000 into AI-assisted teaching more quickly than a school that first needs basic infrastructure, procurement guidance, and responsible-use rules.

2.3 Organisational and pedagogical frameworks

The conceptual framework draws on SELFIE/DigCompOrg, CFIR, TOE, UNESCO AI competency frameworks, TPACK, digital-divide theory, and capability theory to justify a broader index. Broad digital indicators such as STEM, e-learning, coding, robotics, and digital platforms are retained, but they are treated as foundational capacity rather than direct AI implementation. This prevents the index from overclaiming that general digital activity is already AI practice.

3. Data and Methodology

Table 1. Public evidence pipeline

Step	Count	Interpretation
Secondary schools in baseline	441	Official school-level analytical frame
School websites successfully collected	440	Public school website evidence
HTML pages saved	8,231	Website pages and public web records
PDFs saved	1,931	Public PDF documents
Usable documents after text extraction	8,763	Documents with extractable text
Cleaned evidence items	12,997	Deduplicated public evidence snippets
Schools with at least one cleaned evidence item	341	Not a performance count; partly affected by publication visibility

Note. Counts are taken from the processed data files. OCR was not used, so scanned PDFs may be underrepresented.

The evidence cleaning process removes near-duplicate evidence by school, source URL, inferred dimension, inferred sub-dimension, and snippet signature. Evidence is weighted by confidence: high-confidence evidence receives greater weight than medium- or low-confidence evidence. Each school-dimension score is capped on a small ordinal scale before conversion to 0-100, so a long archive cannot dominate a dimension simply by repeating the same keyword many times.

Natural Language Processing (NLP) is used to classify cleaned public-evidence items into conceptual dimensions. Each cleaned evidence item is compared with multilingual natural-language prototypes for the foundational-capacity, early-AI-signal, and AI-relevance dimensions. The classifier uses the sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 model, a multilingual Sentence-BERT model that maps sentences and paragraphs into a dense semantic embedding space for similarity comparison (Reimers & Gurevych, 2019; Sentence-Transformers, n.d.; Wang et al., 2020). In this analysis, 7,778 cleaned items were treated as AI-semantic candidates and the median NLP confidence score was 0.712.

The NLP classifier uses separate similarity thresholds for candidate and non-candidate evidence. Metadata and keyword-derived fields are retained for audit and for deciding whether a lower candidate threshold is allowed, but final dimension assignment comes from semantic similarity to the prototype descriptions. NLP confidence is reported for validation, while the main scoring uses the confidence-weighted evidence design so that evidence confidence remains transparent.

This approach is appropriate for early monitoring, but it is not a substitute for direct school reporting. Website practices differ: some schools publish detailed reports, archived PDFs, and long web pages; others may be active in practice but publish less. For this reason the analysis adds a publication visibility layer and repeatedly uses cautious language such as public evidence, visible signal, candidate for follow-up validation, and potential support need.

4. Index Construction

The index structure uses an early-warning support logic rather than a school-ranking logic. It distinguishes between capacity to absorb future funding, early direct AI signals, quality of AI use, support need, and publication visibility.

Table 2. Index components

Index layer	What it measures	How to interpret it
Foundational AI Implementation Capacity	Visible enabling conditions such as planning, digital infrastructure, teacher development, curriculum integration, student opportunities, governance, and resource mobilisation.	Capacity to absorb future AI funding; not direct proof of AI implementation.
Early AI-Specific Signal	Direct AI-related public evidence such as AI literacy, AI-assisted teaching, AI tools, AI teacher development, AI assessment, responsible AI, cross-subject AI use, and procurement.	Early signal only. Low evidence before rollout is not school failure.
AI Quality of Use	A diagnostic layer focusing on deeper use: teacher development, cross-subject use, responsible AI, AI literacy, assessment, and AI-assisted teaching.	Quality-oriented public signal, not full lesson-level evaluation.
Early-Warning Support Need	A support triage score combining low foundational capacity, teacher-development gap, governance gap, infrastructure gap, and low early AI-specific signal.	Candidate need for validation and support; not a school-quality measure.
Evidence Confidence / Publication Visibility	Document volume, usable text, evidence source count, high-confidence evidence, and dimensions with evidence.	Reliability and publication-practice diagnostic; not school quality.

Table 3. Conceptual dimension map for public evidence coding

Dimension	Why it belongs	Framework support	Public evidence examples
Strategic leadership and planning	AI funding requires planning, not just buying tools.	DigCompOrg; TOE; capability theory	School plan; annual plan; major concerns; AI or digital strategy; development plan.
Foundational digital infrastructure	Schools need basic digital capacity before AI can work.	SELFIE/DigCompOrg; TOE; digital divide theory	E-learning platforms; LMS; Google Classroom; Microsoft Teams; BYOD; tablets; computer rooms; smart classrooms; AI lab.
Teacher development and human capacity	AI implementation depends heavily on teachers' confidence and training.	SELFIE; TPACK; capability theory	Teacher training; professional development; staff workshop; AI teacher training; communities of practice.
Curriculum and pedagogical integration	AI should enter teaching and learning, not only administration or procurement.	TPACK; DigCompOrg teaching and learning practices	STEM; STEAM; coding; robotics; AI literacy; cross-subject learning; project-based learning; digital pedagogy.
Student digital and AI learning opportunities	The policy aim is student learning opportunity, not only school capacity.	Capability theory; digital divide theory	AI literacy activities; competitions; innovation activities; coding teams; robotics teams; digital ambassadors.
Governance, responsible use, and safeguards	AI creates risks around privacy, ethics, academic integrity, and data use.	DigCompOrg; capability theory; responsible AI governance	Privacy; cybersecurity; data security; digital citizenship; responsible AI; AI ethics; academic integrity; acceptable-use policy.
Implementation administration and resource mobilisation	Some schools may have ideas but lack procurement or project-management capacity.	TOE; public administration implementation theory	Procurement; tenders; subscriptions; grant use; vendor or platform selection; implementation reports.

Equity and contextual conversion risk	Equal funding may not become equal capability across schools.	Digital divide theory; capability theory	District income; public rental housing share; fee/remission evidence; NCS support; SEN support; finance type; resource constraints.

Note. The first seven dimensions feed the main school-level public-evidence index. District socioeconomic context is used in Notebook 04 as a sensitivity layer because it is district-level, not school-level.

The main Early-Warning Support Need formula uses five observable school-evidence components: 25% low foundational capacity, 20% teacher development gap, 15% governance/responsible-use gap, 15% infrastructure gap, and 10% low early AI-specific signal. These add to 85% of the suggested full model and are rescaled to 100% in the no-context specification. A separate context-sensitivity specification includes a 15% district contextual need component.

Important interpretation rule

The support-need score is not a grade. It is a triage signal. A school may have low visible public evidence because it publishes little online, because activities are not yet underway, or because it needs support. The appropriate administrative response is validation and support, not labelling.

5. Findings

5.1 Headline distributions: capacity and direct AI signals remain low and uneven

Table 4. Overall distribution of monitoring indices

Metric	Mean	Median	P25	P75	P90	Gini
Foundational AI Implementation Capacity	31.01	23.81	4.76	47.62	85.71	0.540
Early AI-Specific Signal	20.91	12.50	0.00	33.33	62.50	0.613
AI Quality of Use	20.42	11.11	0.00	33.33	61.11	0.631
Early-Warning Support Need	74.28	84.94	61.13	98.60	100.00	0.204
Evidence Confidence / Publication Visibility	50.11	53.78	26.34	73.13	85.07	0.304

Note. All index values are on a 0-100 scale. Higher support need means stronger candidate need for validation and support, not lower school quality.

Mean foundational capacity is 31.01/100, and the median is 23.81/100. Mean early AI-specific signal is only 20.91/100, with a median of 12.50/100. Quality of use is similarly thin, with a mean of 20.42/100 and median of 11.11/100. The Gini values for early AI signal (0.613) and quality (0.631) indicate substantial unevenness in visible direct AI evidence.

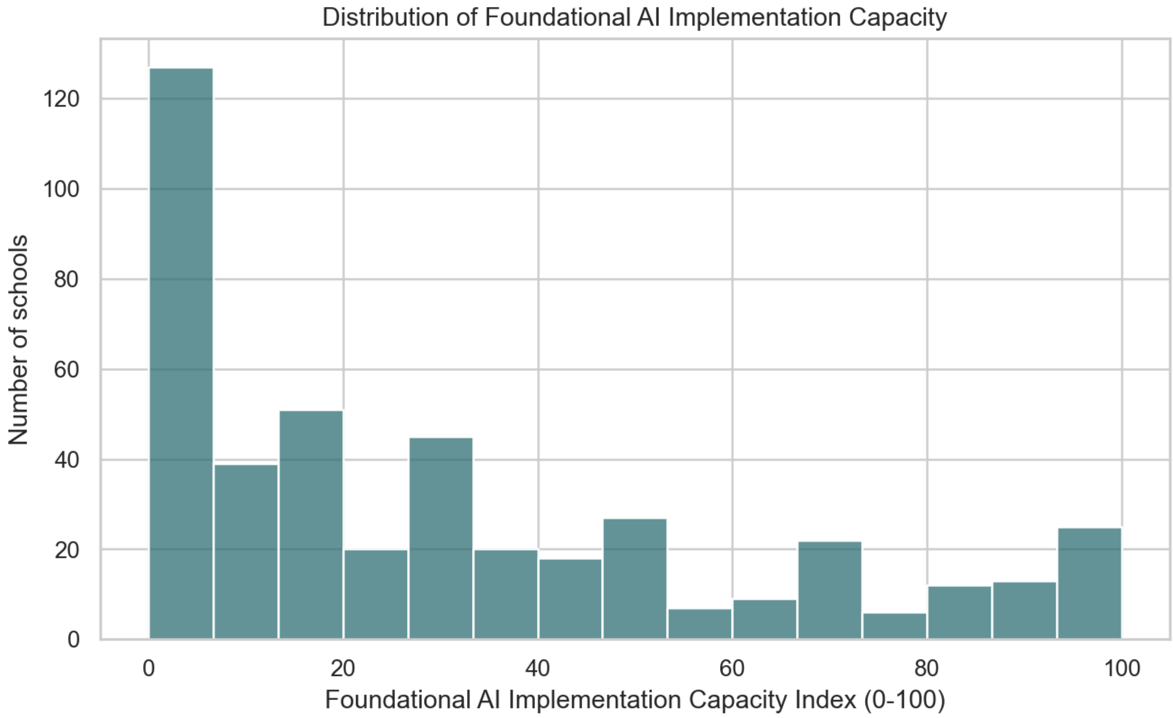

Figure 1. Distribution of Foundational AI Implementation Capacity scores.

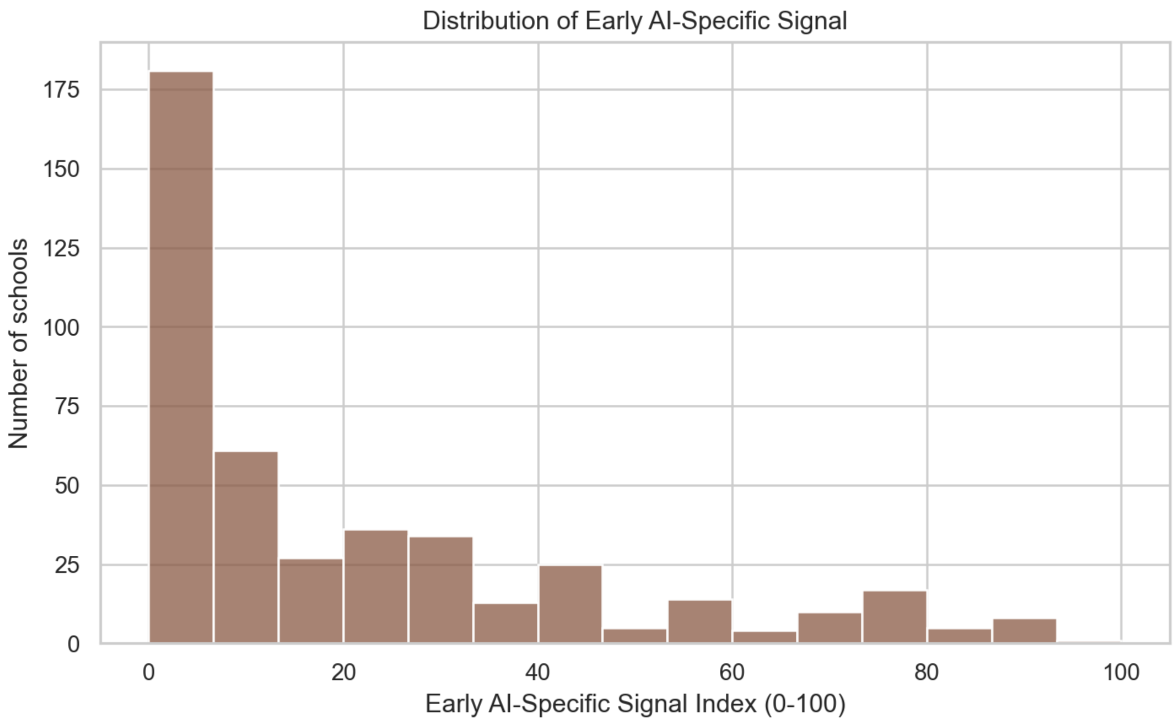

Figure 2. Distribution of Early AI-Specific Signal scores.

5.2 Foundational capacity is most visible in planning, student opportunity, and infrastructure signals

Table 5. Foundational capacity dimensions

Dimension	Schools with evidence (%)	Mean score	Evidence items
Strategic leadership and planning	61.22	46.03	2,878
Student digital / AI learning opportunities	59.64	46.26	3,449

Foundational digital infrastructure	57.60	40.21	2,371
Curriculum and pedagogical integration	48.53	33.18	2,024
Teacher development and human capacity	29.02	19.95	1,191
Administrative / resource mobilisation capacity	28.12	18.14	640
Governance and responsible use	24.04	13.30	390

Under NLP coding, strategic leadership and planning is the most visible foundational dimension: 61.22% of schools have evidence, with a mean score of 46.03. Student digital or AI learning opportunities are also visible at 59.64%, and foundational digital infrastructure appears in 57.60% of schools. Weaker visible areas are governance and responsible use (24.04%, mean 13.30), administrative/resource mobilisation (28.12%, mean 18.14), and teacher development capacity (29.02%, mean 19.95).

This pattern matters because the policy does not only require enthusiasm for AI. It requires schools to plan, train teachers, procure or manage tools, establish responsible-use routines, and integrate AI into teaching and learning. The public evidence suggests that the governance-heavy and implementation-administration parts of capacity are less visible than broad curriculum or innovation signals.

5.3 Direct AI-specific evidence is still thin

Table 6. Early AI-specific signal dimensions

Dimension	Schools with evidence (%)	Mean score	Evidence items
AI procurement	50.34	35.45	2,723
AI literacy activities	46.71	28.34	1,043
Cross-subject AI use	45.12	26.98	1,052
AI-assisted teaching	39.00	26.61	1,515
AI assessment	32.20	19.50	645
AI teacher development	25.62	14.06	472
AI tools / platforms	18.14	9.30	170
Responsible AI	14.74	7.03	155

The NLP classification produces a direct AI profile in which AI procurement is the most visible dimension, with evidence in 50.34% of schools and a mean score of 35.45. AI literacy activities appear in 46.71% of schools, cross-subject AI use in 45.12%, and AI-assisted teaching in 39.00%. The thinnest direct AI evidence remains responsible AI (14.74%), AI tools or platforms (18.14%), AI teacher development (25.62%), and AI assessment (32.20%).

This does not mean schools have failed to implement AI. The policy period is early, and public evidence is incomplete. The appropriate interpretation is that direct AI-specific signals are still thin and uneven, so schools may need differentiated support before being expected to convert grants into multi-subject AI-assisted teaching.

5.4 Percentile-based support tiers avoid labelling hundreds of schools as high risk

Table 7. Percentile-based support-prioritisation tiers

Tier	Meaning	Schools	Mean support need	Mean capacity	Mean early AI signal	Mean visibility
Tier 1	Tier 1: Immediate validation / intensive support candidate	45	100.00	0.00	0.00	15.22
Tier 2	Tier 2: Targeted capacity-building candidate	66	99.79	0.72	0.00	15.43
Tier 3	Tier 3: General monitoring / standard support	220	83.30	24.03	13.77	51.75
Tier 4	Tier 4: Lower immediate support need / peer-learning candidate	110	30.41	75.84	56.29	81.92

Note. Tier 1 is the top 10% of support need, Tier 2 the next 15%, Tier 3 the middle 50%, and Tier 4 the lowest 25%.

The new tiering rule is deliberately percentile-based. It avoids the earlier problem of labelling a very large majority of schools as very high risk. Tier 1 contains 45 schools, Tier 2 contains 66, Tier 3 contains 220, and Tier 4 contains 110. Tier 1 and Tier 2 should be interpreted as candidates for validation and support, not as a public list of poor performance.

The capacity-signal scatterplot gives a complementary diagnostic view. Using median thresholds, 208 schools show higher foundational capacity and higher early AI-specific signal; 16 show higher capacity but lower early AI-specific signal; 19 show lower foundational capacity but higher early AI-specific signal; and 198 show lower public evidence on both axes.

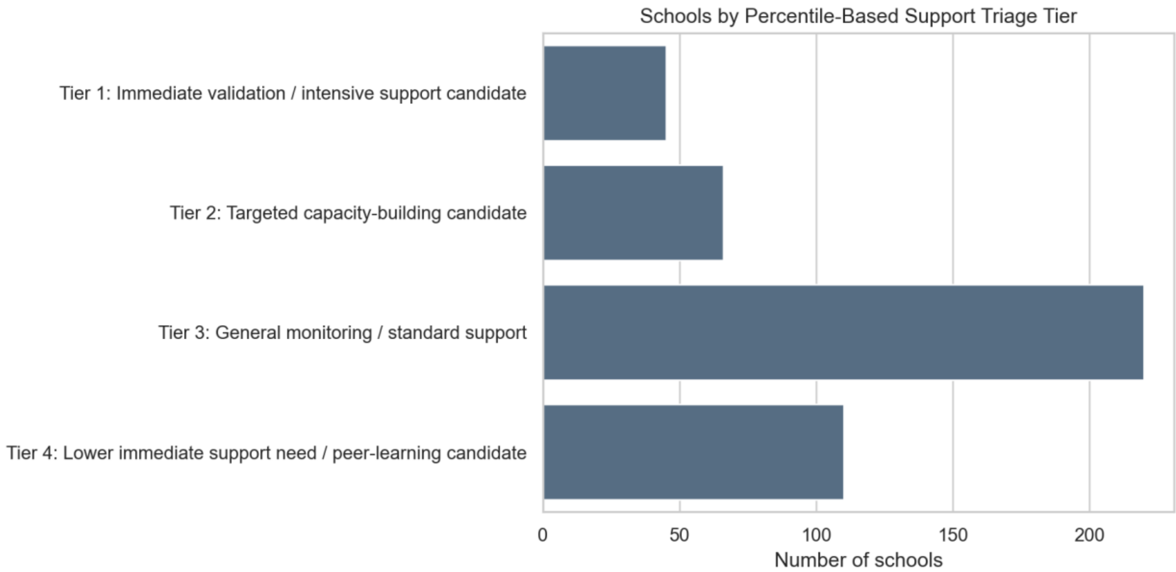

Figure 3. Schools by percentile-based support triage tier.

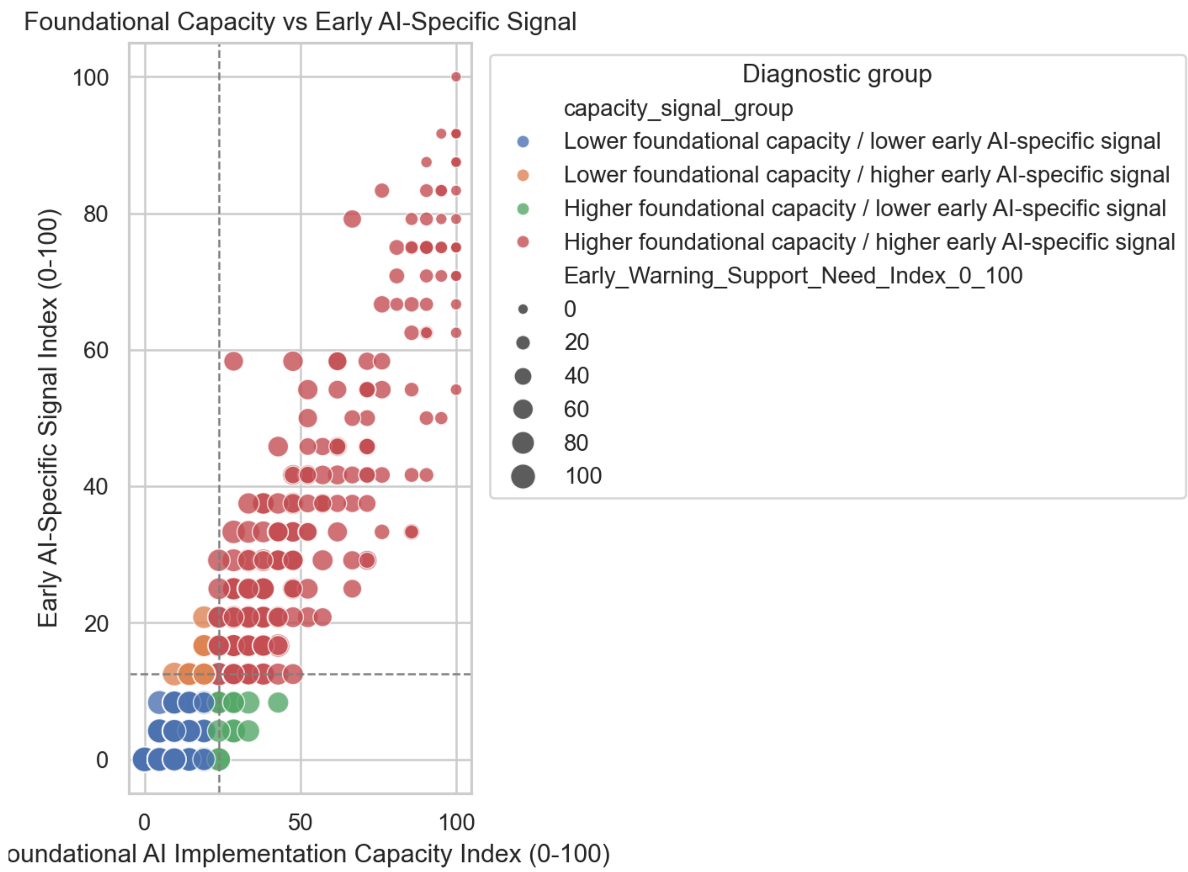

Figure 4. Foundational capacity versus early AI-specific signal.

5.5 District patterns are visible, but district is not the cause

Table 8. District support-need patterns

Category	District	Schools	Mean capacity	Mean early AI	Mean support need	Tier 1/2 share
Highest mean support need	North	20	15.48	8.12	88.81	40.0%
Highest mean support need	Islands	9	15.87	13.89	85.20	55.6%
Highest mean support need	Central & Western	11	23.38	12.50	82.74	27.3%
Highest mean support need	Kwai Tsing	31	27.50	17.34	78.42	25.8%
Highest mean support need	Tai Po	20	27.86	16.04	77.27	20.0%
Lowest mean support need	Wong Tai Sin	22	40.04	28.03	66.30	22.7%
Lowest mean support need	Eastern	30	36.98	24.58	68.45	30.0%
Lowest mean support need	Yuen Long	39	35.41	23.40	69.54	20.5%
Lowest mean support need	Kwun Tong	33	35.50	27.91	69.64	21.2%
Lowest mean support need	Sai Kung	26	35.53	25.64	69.64	26.9%

Note. District comparisons are descriptive. They should be interpreted with school-level validation and publication-visibility checks.

Mean support need is highest in North (88.81), Islands (85.20), Central & Western (82.74), Kwai Tsing (78.42), Tai Po (77.27). It is lowest in Wong Tai Sin (66.30), Eastern (68.45), Yuen Long (69.54), Kwun Tong (69.64), Sai Kung (69.64). The highest-lowest district gap in mean support need is 22.51 points. This is a descriptive pattern, not proof that district itself causes implementation capacity.

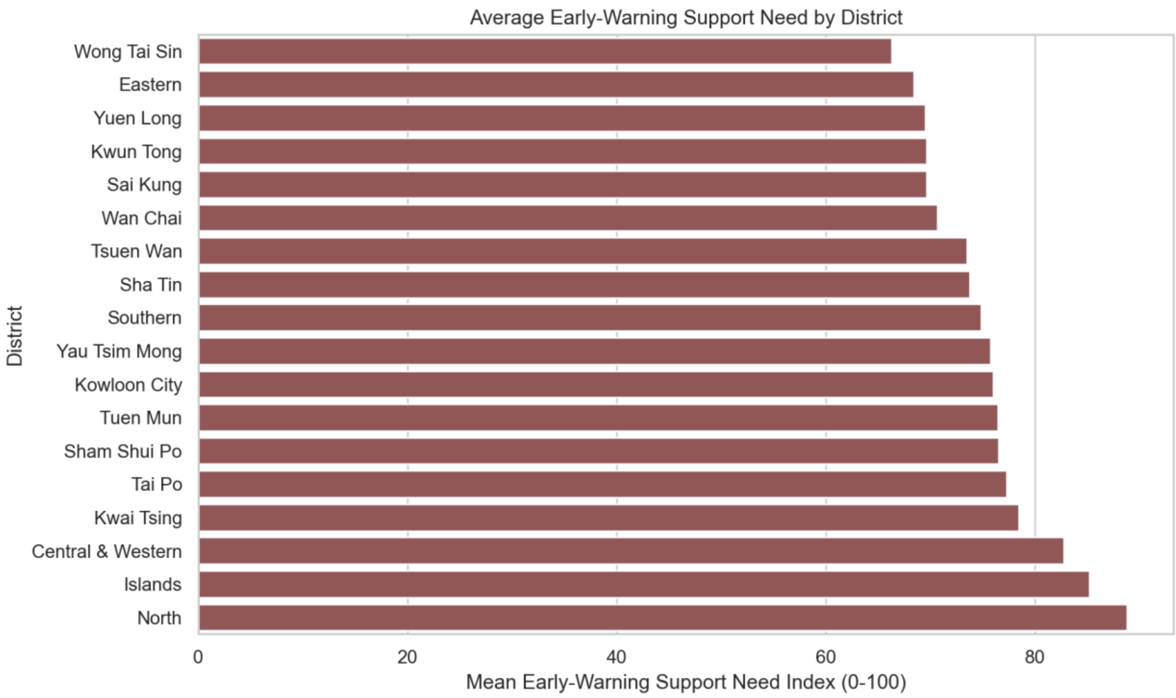

Figure 5. Average Early-Warning Support Need by district.

The district socioeconomic context is included only as a sensitivity layer. District-level census variables are useful background information, but they do not describe individual schools. When contextual need is included, district-level socioeconomic indicators correlate more strongly with the context-sensitive support score. For example, median monthly employment earnings correlates negatively with mean context-sensitive support need (Spearman rho = -0.616), while the district contextual need index correlates positively (rho = 0.620).

5.6 Publication visibility is a major trust issue

Table 9. Publication visibility correlations

Publication variable	Outcome	Spearman rho
Publication visibility	Foundational capacity	0.964
Publication visibility	Early AI-specific signal	0.931
Publication visibility	AI quality of use	0.904
Publication visibility	Early-warning support need	-0.941
Total extracted text length	Foundational capacity	0.889
Total extracted text length	Early-warning support need	-0.868
Publication volume index	Foundational capacity	0.780
Publication volume index	Early-warning support need	-0.760

Note. All listed correlations are statistically significant in the publication-visibility analysis. They show association, not causality.

Publication visibility is strongly associated with the monitoring scores. The visibility index correlates 0.964 with foundational capacity, 0.931 with early AI-specific signal, 0.904 with quality, and -0.941 with support need. Total extracted text length has similar associations (0.889 with foundational capacity and -0.868 with support need). This means the index partly captures whether schools publish evidence publicly. That is not a defect if the tool is used for triage, but it would be a serious problem if it were used as a public league table.

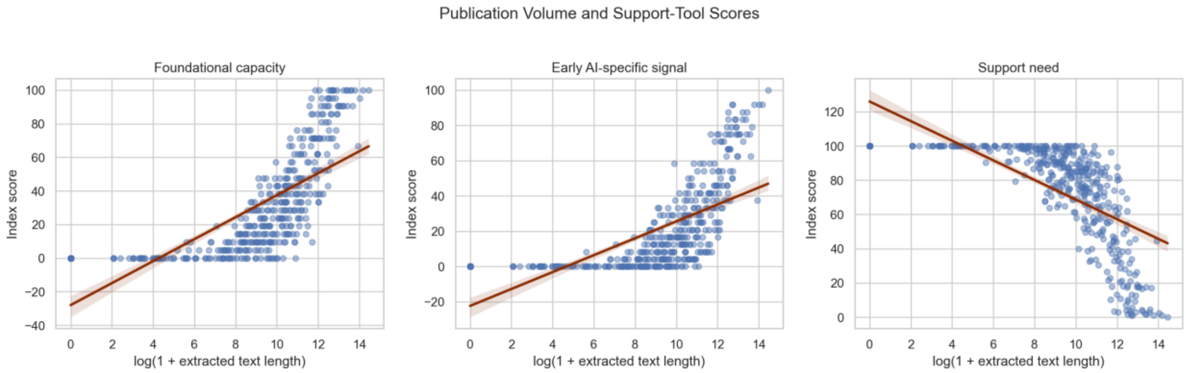

Figure 6. Publication volume and support-tool scores.

Publication-adjusted district summaries change the district ordering. For example, North, Islands have the highest unadjusted support-need means, while after publication adjustment the highest district means are Central & Western (82.03), Yau Tsim Mong (79.73), Tsuen Wan (79.51). This reinforces the report's central caution: public website monitoring is useful, but incomplete.

5.7 School characteristics matter, but they do not replace direct evidence

Table 10. Selected school-characteristic summaries

Characteristic	Group	Schools	Mean capacity	Mean early AI	Mean support need	Tier 1 share
School Type	Aided	354	31.15	20.80	74.12	9.9%
School Type	CAPUT	2	2.38	2.08	99.06	50.0%
School Type	DSS	56	24.57	17.71	80.01	16.1%
School Type	Gov't	29	43.68	29.74	63.51	0.0%
Finance Type	AIDED	340	31.06	20.79	74.18	10.3%
Finance Type	CAPUT	2	2.38	2.08	99.06	50.0%

Finance Type	DIRECT SUBSIDY SCHEME	54	23.55	16.90	80.99	16.7%
Finance Type	GOVERNMENT	27	44.62	31.33	62.35	0.0%
Finance Type	Unknown	18	35.19	21.76	71.19	0.0%

Note. Group comparisons are descriptive and should not be treated as causal explanations.

The group-comparison tests find statistically detectable differences across school types for early-warning support need and foundational capacity. However, these are group-level patterns. They should not be used to infer individual school need without validation. The analysis also shows that publication visibility and evidence volume are among the strongest correlates of the index outcomes, which is another reason to rely on manual checking before making support decisions.

6. Discussion

6.1 Main interpretation: a support-conversion problem

The evidence supports a clear public administration interpretation. Hong Kong's AI education policy is not only a funding question; it is a support-conversion question. Schools differ in visible capacity to convert money, guidance, and policy expectations into teacher practice and student opportunity. Equal funding can therefore produce unequal implementation unless support is matched to starting capacity.

6.2 Why the report should avoid ranking language

The index should not be described as a ranking, grading tool, or proof of poor performance. A low public signal can mean several things: low implementation, unpublished implementation, sparse website publication, poor extractability of documents, or evidence that requires manual review. The phrase Early-Warning Support Need is therefore more accurate than Equity Risk. It keeps the focus on administrative response: validation, capacity-building, teacher support, governance support, and follow-up.

6.3 Why publication visibility should remain visible

The Evidence Confidence / Publication Visibility layer is essential. It helps prevent website publication practice from being confused with school quality. Schools with richer websites and longer archives are more likely to show public evidence. Schools with lower visibility should not automatically be assumed to have lower actual practice. Instead, low visibility plus high support need should trigger follow-up validation.

7. Policy Recommendations

1. Use the tiers as support pathways, not as public rankings. Tier 1 schools should be candidates for immediate validation and intensive support; Tier 2 for targeted capacity-building; Tier 3 for general monitoring and standard support; and Tier 4 for lower immediate support need or voluntary peer learning.
2. Pair funding with basic implementation scaffolds. Provide planning templates, procurement guidance, sample responsible-use policies, teacher development sequences, and subject-level AI-assisted teaching examples.
3. Create a small standardised reporting requirement for funded schools. Useful indicators include subjects using AI-assisted teaching, teacher training completed, teaching resources developed, student AI literacy activities, procurement choices, and responsible-AI governance arrangements.
4. Validate before acting. Use public evidence to shortlist candidates for follow-up, then verify through school self-report, document review, or sampling. Low publication visibility should increase the need for validation, not blame.
5. Invest in governance and responsible use. The low visibility of governance, responsible AI, procurement, and infrastructure signals suggests that schools need practical guidance on privacy, data security, academic integrity, bias, assessment integrity, age-appropriate use, and teacher oversight.
6. Use peer learning carefully. Tier 4 schools or high-capacity/high-signal schools may be invited into voluntary demonstration, mentoring, and resource-sharing roles. This should not become a prestige ranking or reputational pressure mechanism.

8. Limitations and Validation Needs

- Website evidence is incomplete. Absence of public evidence does not mean absence of activity.

- Publication practices vary strongly across schools. The visibility correlations show that document volume and text length shape observed scores.
- NLP-classified public evidence requires manual validation. Even after deduplication, semantic classification, and confidence weighting, snippets can still be weak, duplicated in meaning, or context-dependent.
- District socioeconomic context is district-level. It should not be interpreted as school-level student socioeconomic composition.
- The index predates full rollout of the funding programme. Direct AI-specific evidence should therefore be read as early signal, not implementation failure.
- Machine learning remains exploratory. The models can find some patterns, but sample size is small and model targets are constructed from public evidence. Feature importance is not causality and should not guide high-stakes decisions.

The most important next step is manual validation. A reasonable validation design would sample schools from all four support tiers and from low-visibility/high-support-need cases. Reviewers should check whether evidence snippets genuinely indicate the coded dimension, whether the school website is unusually sparse or unusually rich, and whether future EDB implementation reports confirm or contradict the public evidence.

9. Conclusion

The analysis strengthens the project's public administration contribution. The key finding is not that some schools should be ranked above others. The key finding is that publicly observable implementation capacity is low, uneven, and deeply affected by publication visibility. AI education policy will be more equitable if funding is paired with tiered support, validation, teacher development, governance guidance, and practical reporting.

AI for All therefore requires Capacity for All. The purpose of the index is to identify where implementation support may need to arrive first, so that AI education funding does not quietly become AI for already-ready schools.

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Appendix: Output Tables and Figures Used

Appendix Table A1. Main generated outputs used in this report

Output	Use in report
Notebook 03	NLP dimension classification, index construction, evidence cleaning, dimension scoring, support tiers, and index distributions.
Notebook 04	District patterns, publication-visibility checks, sensitivity tests, school-characteristic summaries, and exploratory ML diagnostics.
secondary_school_ai_indices_0_100_nlp.csv	School-level monitoring indices generated from NLP-classified evidence.
secondary_school_support_priority_groups_nlp.csv	Percentile support tiers and diagnostic capacity-signal groups generated from NLP-classified evidence.
support_tool_distribution_statistics.csv	Headline means, medians, percentiles, and inequality statistics.
foundational_capacity_dimension_summary.csv	Foundational capacity dimension frequencies and mean scores.
early_ai_specific_signal_dimension_summary.csv	Direct AI-specific dimension frequencies and mean scores.
district_support_need_summary.csv	District descriptive summaries and Tier 1/2 shares.
publication_visibility_correlations.csv	Publication-bias and trust diagnostics.
support_triage_sensitivity_robustness.csv	Robustness of percentile tiering under context and strict-confidence variants.

Appendix Table A2. Support-tier sensitivity checks

Variant	Tier 1 schools	Tier 1 share	Agreement with main	Tier 1 overlap with main
district_context_sensitivity_percentile_tiers	45	10.2%	0.762	46.7%
main_no_context_percentile_tiers	45	10.2%	1.000	100.0%
strict_confidence_weights_percentile_tiers	45	10.2%	0.918	86.7%

Note. The strict-confidence variant preserves most Tier 1 cases; the district-context version changes more cases because it adds a district-level contextual need layer.

Appendix Table A3. Exploratory machine-learning diagnostics

Model	Target	Holdout R2	Balanced accuracy	Holdout ROC-AUC	CV metric
RandomForestRegressor	Early_Warning_Support_Need_Index_0_100	-0.062			-0.106
RandomForestClassifier	tier_1_support_priority		0.540	0.661	0.642

Note. These models are exploratory diagnostics only. They should not be used for high-stakes school-level targeting.